In …

```
(* Settings *)
SetOptions[
    Plot,
    Background -> White,
    AxesStyle -> Black,
    LabelStyle -> Black,
    TicksStyle -> Black
];

SetOptions[
    Plot3D,
    Background -> White,
    BoxStyle -> Black,
    AxesStyle -> Black,
    LabelStyle -> Black,
    TicksStyle -> Black
];

SetOptions[
    ParametricPlot,
    Background -> White,
    AxesStyle -> Black,
    LabelStyle -> Black,
    TicksStyle -> Black
];
```

Practical-1

Solution of Cauchy problem for First-Order Partial Differential Equation (PDE)

Problem-1: To obtain the solution of the linear equation

$$u_x - u_y = 1$$

with the Cauchy data $u(x, 0) = x^2$

Solution:

In …

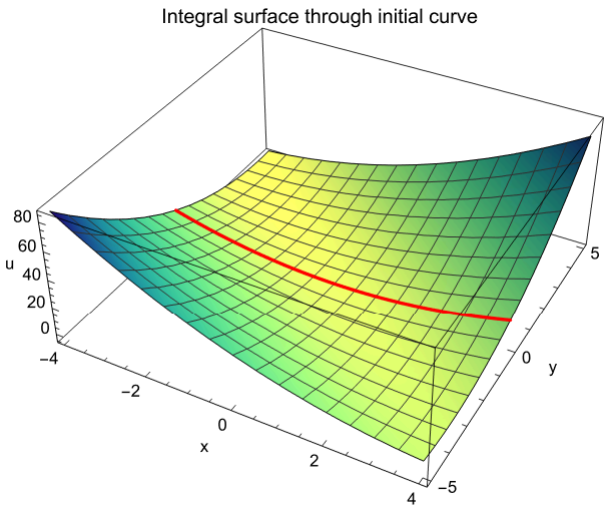
```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] - D[u[x, y], y] == 1

initialCondition = u[x, 0] == x^2;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    ColorFunction -> (ColorData["BlueGreenYellow"][1 - #3] &),
    ColorFunctionScaling -> True
];

initialCurve = ParametricPlot3D[
    {t, 0, t^2}, {t, -4, 4},
    PlotStyle -> {Thick, Red}
];
```

```
Show[surface, initialCurve]
```

Out…

$-u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 1$

$\{\{u[x, y] \to x^2 - y + 2\,x\,y + y^2\}\}$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-2: To obtain the solution of the linear equation

$$u(x+y)\,u_x + u(x-y)\,u_y = x^2 + y^2$$

with the Cauchy data $u(x, 2x) = 0$

Solution:

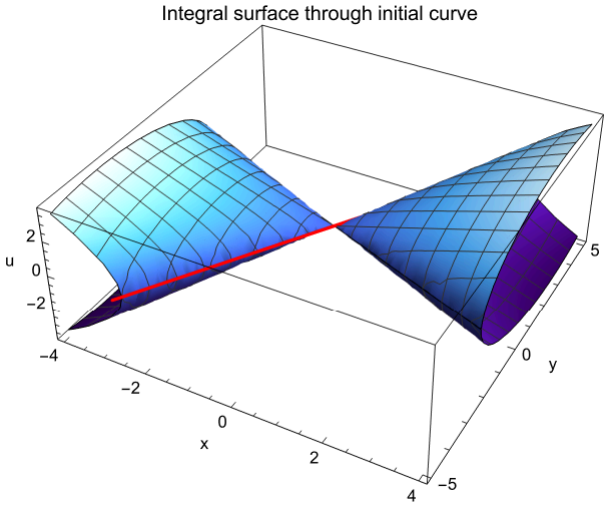
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = u[x, y] (x + y) D[u[x, y], x] + u[x, y] (x - y) D[u[x, y], y] == x^2 + y^2

initialCondition = u[x, 2 x] == 0;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    ColorFunction -> (ColorData["DeepSeaColors"]),
    ColorFunctionScaling -> True
];

initialCurve = ParametricPlot3D[
    {t, 2*t, 0}, {t, -4, 4},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

$$(x-y)\,u[x,y]\,u^{(0,1)}[x,y]+(x+y)\,u[x,y]\,u^{(1,0)}[x,y]==x^2+y^2$$

$$\left\{\left\{u[x,y]\to-\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\},\left\{u[x,y]\to\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\},\left\{u[x,y]\to-\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\},\left\{u[x,y]\to\sqrt{\frac{2}{7}}\sqrt{2x^2+3xy-2y^2}\right\}\right\}$$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-3: To obtain the solution of the linear equation

$$u_x + u_y = u^2$$

(a) with the Cauchy data $x + y = 0$, $u = 1$
(b) with the Cauchy data $u(x, 0) = \tanh(x)$

Solution:

(a)

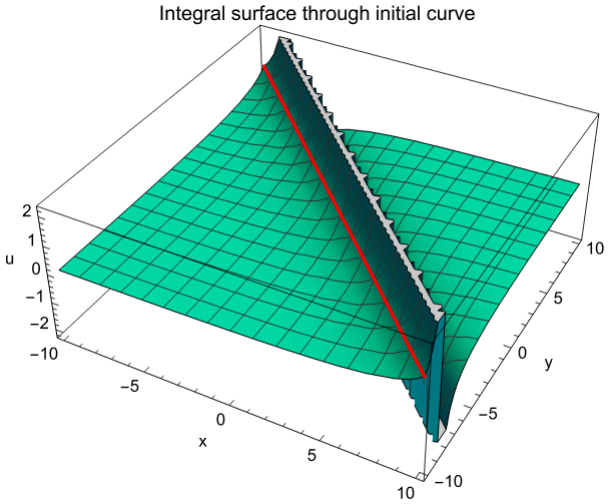
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + D[u[x, y], y] == (u[x, y])^2

(* x + y = 0, u = 1 => y = -x, u = 1 => u[x, -x] = 1 *)
initialCondition = u[x, -x] == 1;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -10, 10}, {y, -10, 10},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    PlotStyle -> Cyan
];

initialCurve = ParametricPlot3D[
    {t, -t, 1}, {t, -10, 10},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

$$u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == u[x, y]^2$$

$$\left\{\left\{u[x, y] \to -\frac{2}{-2 + x + y}\right\}\right\}$$



(b)

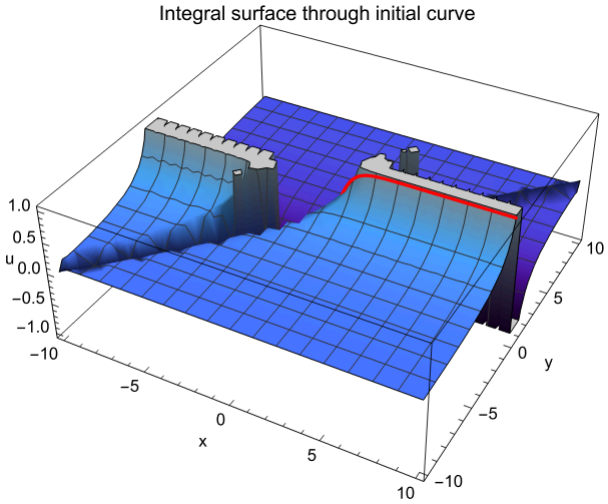
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + D[u[x, y], y] == (u[x, y])^2

initialCondition = u[x, 0] == Tanh[x];

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

surface = Plot3D[
    u[x, y] /. solution, {x, -10, 10}, {y, -10, 10},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface through initial curve",
    ColorFunction -> (ColorData["DeepSeaColors"]),
    ColorFunctionScaling -> True
];

initialCurve = ParametricPlot3D[
    {t, 0, Tanh[t]}, {t, -10, 10},
    PlotStyle -> {Thick, Red}
];

Show[surface, initialCurve]
```

Out…

```
u^(0,1)[x, y] + u^(1,0)[x, y] == u[x, y]^2
```

$$\left\{\left\{u[x, y] \to \frac{1}{-y + \text{Coth}[x - y]}\right\}\right\}$$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Practical-2

Plotting the Characteristics for the First-Order PDE

Problem-1: To find the characteristics of the equation

$$(u - y)\, u_x + y\, u_y = x + y$$

and plot them.

Solution:

The auxiliary equations of the given PDE is

$$\frac{dx}{u - y} = \frac{dy}{y} = \frac{du}{x + y} \tag{1}$$

We have,

$$\Rightarrow \frac{dx + du}{(u - y) + (x + y)} = \frac{dy}{y}$$

$$\Rightarrow \frac{d(x + u)}{x + u} = \frac{dy}{y}$$

$$\Rightarrow \int \frac{d(x + u)}{x + u} = \int \frac{dy}{y}$$

$$\Rightarrow \log(x + u) = \log(y) + \log(c_1)$$

$$\Rightarrow \frac{x + u}{y} = c_1$$

$$\therefore\ u = c_1\, y - x \tag{2}$$

Here, c_1 is a constant.

Again, we have from Eq.(1),

$$\Rightarrow \frac{dx + dy}{(u - y) + y} = \frac{du}{x + y}$$

$$\Rightarrow \frac{d(x + y)}{u} = \frac{du}{x + y}$$

$$\Rightarrow (x + y)\, d(x + y) = u\, du$$

$$\Rightarrow \int (x + y)\, d(x + y) = \int u\, du$$

$$\Rightarrow \frac{(x + y)^2}{2} = \frac{u^2}{2} + \frac{c_2}{2}$$

$$\Rightarrow (x + y)^2 - u^2 = c_2$$

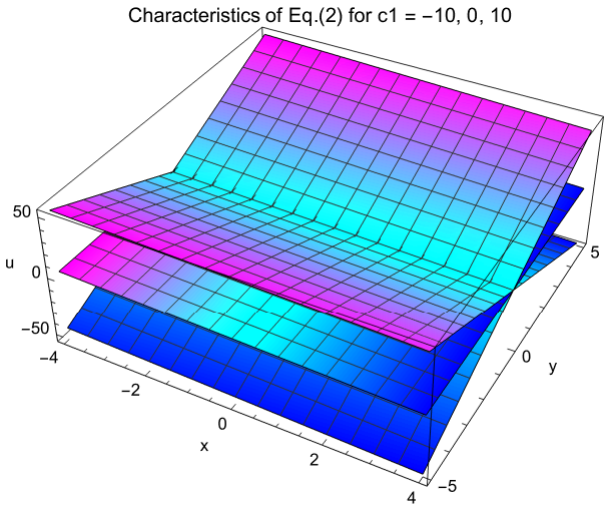
$$\therefore\ u = \sqrt{(x + y)^2 - c_2} \tag{3}$$

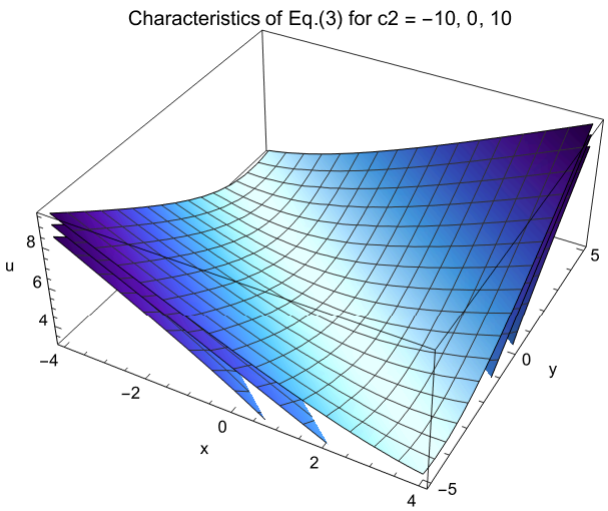
Here, c_2 is a constant.

In …

```
Clear[f, h]

(* c = c1 = -10, 0, 10 *)
f = Table[
   Plot3D[c y - x, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Characteristics of Eq.(2) for c1 = -10, 0, 10",
    ColorFunction -> (Blend[{Blue, Cyan, Magenta}, #3] &),
    ColorFunctionScaling -> True
   ],
   {c, {-10, 0, 10}}
];

Show[f]

(* d = c2 = -10, 0, 10 *)
h = Table[
   Plot3D[Sqrt[(x + y)^2 - d], {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Characteristics of Eq.(3) for c2 = -10, 0, 10",
    ColorFunction -> (ColorData["DeepSeaColors"][1 - #3] &),
    ColorFunctionScaling -> True
   ],
   {d, {-10, 0, 10}}
];

Show[h]
```

Out…





Result: The characteristics of the given PDE are plotted in the output.

Practical-3

Plotting the Integral Surfaces of a given First-Order PDE with Initial Data

Problem-1: To find the solution of the equation

$$u_x + y\,u_y = 0$$

with the Cauchy data $u(0, y) = 4y$

Plot the integral surface with in the range $(x, -4, 4)$ & $(y, -5, 5)$

Solution:

In …

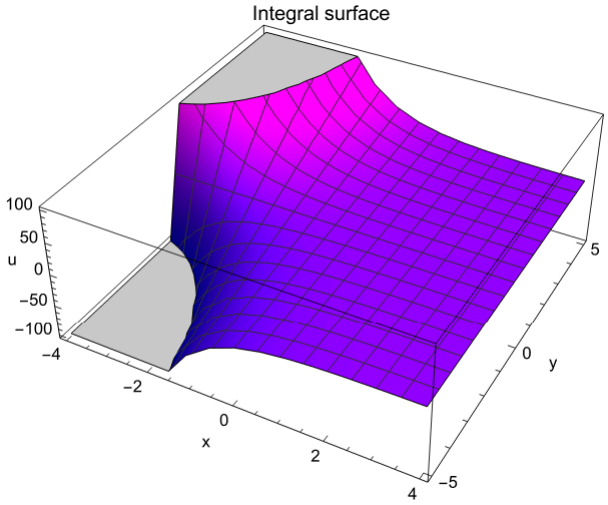
```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + y D[u[x, y], y] == 0

initialCondition = u[0, y] == 4 y;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
```

```
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface",
    ColorFunction -> (Blend[{Blue, Magenta}, #3] &),
    ColorFunctionScaling -> True
]
```

Out…

$y\,u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 0$

$\{\{u[x, y] \to 4\,e^{-x}\,y\}\}$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-2: To obtain the solution of the equation

$$u_x + u\,u_y = 1$$

with the Cauchy data $u(0, y) = 4y$

Plot the integral surface with in the range $(x, -4, 4)$ & $(y, -5, 5)$

Solution:

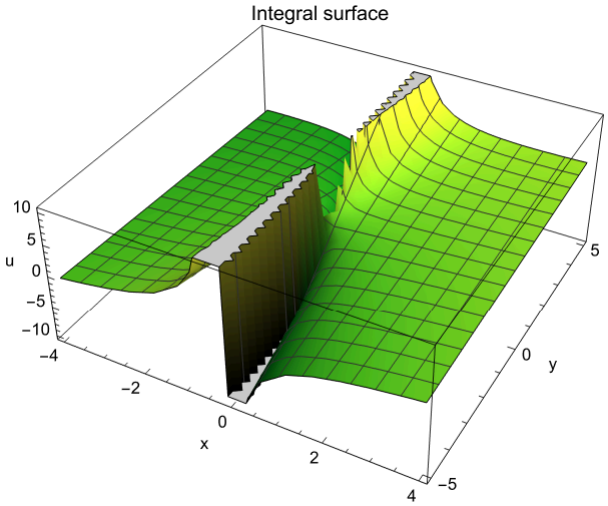
In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + u[x, y] D[u[x, y], y] == 1

initialCondition = u[0, y] == 4 y;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface",
    ColorFunction -> "AvocadoColors",
    ColorFunctionScaling -> True
]
```

Out…

$$u[x, y]\, u^{(0,1)}[x, y] + u^{(1,0)}[x, y] == 1$$

$$\left\{\left\{u[x, y] \to \frac{x + 2x^2 + 4y}{1 + 4x}\right\}\right\}$$



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Problem-3: To obtain the solution of the equation

$$u_x + u_y = 1$$

with the Cauchy data $u(x, 2x) = x^3$

Plot the integral surface with in the range $(x, -4, 4)$ & $(y, -5, 5)$

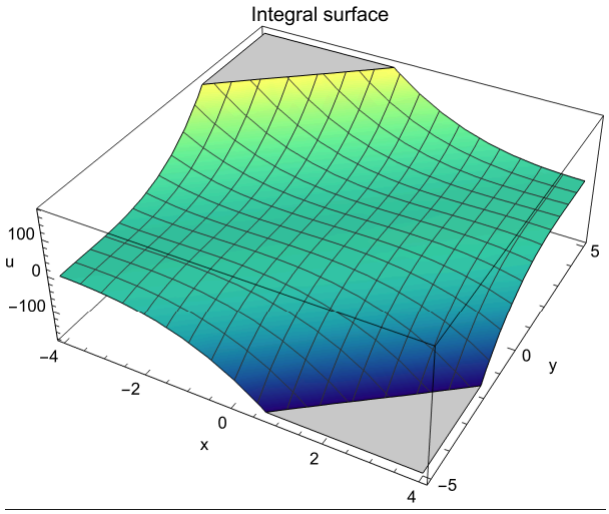
Solution:

In …

```
Clear[x, y, u, t,
    pde, initialCondition,
    solution,
    surface, initialCurve
]

pde = D[u[x, y], x] + D[u[x, y], y] == 1

initialCondition = u[x, 2 x] == x^3;

solution = DSolve[{pde, initialCondition}, u[x, y], {x, y}]

Plot3D[
    u[x, y] /. solution, {x, -4, 4}, {y, -5, 5},
    AxesLabel -> {"x", "y", "u"},
    PlotLabel -> "Integral surface",
    ColorFunction -> (ColorData["BlueGreenYellow"][#3] &),
    ColorFunctionScaling -> True
]
```

Out…

```
u^(0,1)[x, y] + u^(1,0)[x, y] == 1
{{u[x, y] → 2 x - x^3 - y + 3 x^2 y - 3 x y^2 + y^3}}
```



Result: The solution of the Cauchy problem for the given PDE is obtained in the output and visualized as an integral surface.

Practical-4

Solutions of Wave equation for associated conditions

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

(a) $u(x,0) = \phi(x)$, $u(x,0) = \psi(x)$; $x \in \mathbb{R}, t > 0$
(b) $u(x,0) = \phi(x)$, $u_t(x,0) = \psi(x)$, $u(0,t) = 0$; $x > 0, t > 0$
(c) $u(x,0) = \phi(x)$, $u_t(x,0) = \psi(x)$, $u_x(0,t) = 0$; $x > 0, t > 0$
(d) $u(x,0) = \phi(x)$, $u_t(x,0) = \psi(x)$, $u(0,t) = 0$, $u(l,t) = 0$; $x > 0, t > 0$

Problem-1: To obtain the solution of the wave equation

$$u_{tt} - u_{xx} = 0, \quad 0 < x < 1, \quad t > 0$$

for the initial conditions

$$u(x,0) = \sin(x), \quad u_t(x,0) = \cos(x), \quad 0 \le x \le 1, \quad t \ge 0$$

Solution:

In …

```
Clear[x, t, u, pdeWithConditions, solution]

pdeWithConditions = {D[u[x, t], {t, 2}] - D[u[x, t], {x, 2}] == 0,
    u[x, 0] == Sin[x],
    Derivative[0, 1][u][x, 0] == Cos[x]
}

solution = NDSolve[pdeWithConditions, u[x, t], {x, 0, 1}, {t, 0, 4},
    PrecisionGoal -> 3
]

Plot3D[u[x, t] /. solution, {x, 0, 1}, {t, 0, 4},
    AxesLabel -> {"x", "t", "u"},
    PlotLabel -> "Solution surface of the given Wave equation",
    ColorFunction -> (Blend[{Orange, Magenta}, #3] &),
```

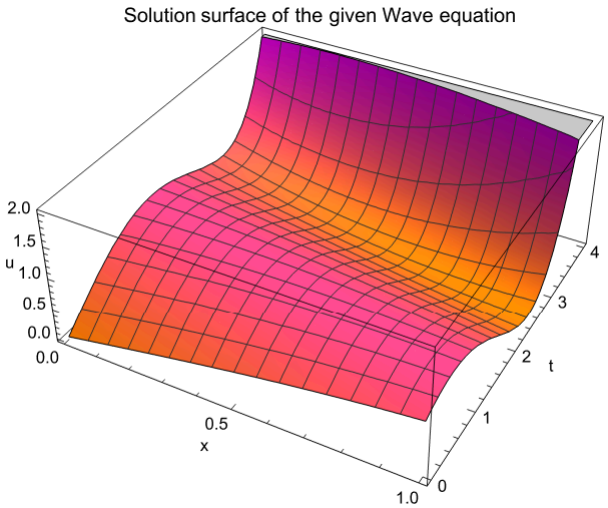
```
    ColorFunctionScaling -> True
]
```

```
NDSolve::bcart: Warning: an insufficient number of boundary conditions have been specified for t
he direction of independent variable x. Artificial boundary effects may be present in the soluti
on.
```

Out…

```
{u^(0,2)[x, t] - u^(2,0)[x, t] == 0, u[x, 0] == Sin[x], u^(0,1)[x, 0] == Cos[x]}
```

```
{{u[x, t] → InterpolatingFunction[ Domain: {{0., 1.}, {0., 4.}} Output: scalar ][x, t]}}
```



Result: The solution of the given Wave equation is obtained and visualized as a surface.

Problem-2: To obtain the solution of the wave equation

$$u_{tt} - u_{xx} = -9.80665, \quad 0 < x < 1, \quad t > 0$$

for the initial conditions

$$u(x,0) = 10\,x^2\,(1-x)^2, \quad u_t(x,0) = 0, \quad u(0,t) = u(1,t) = 0, \quad 0 \le x \le 1, \quad t \ge 0$$

Solution:

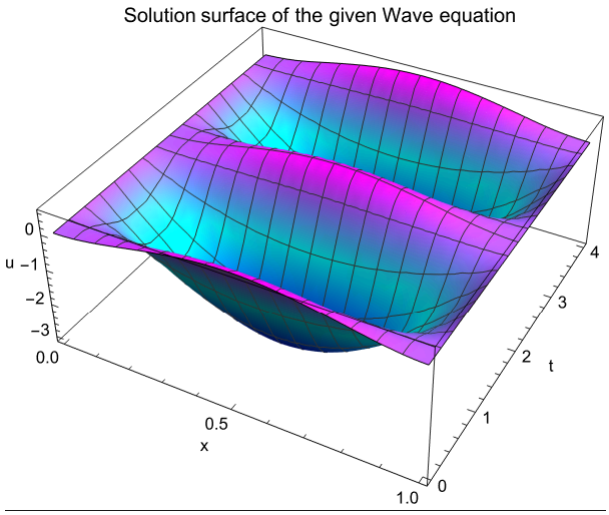
In …

```
Clear[x, t, u, pdeWithConditions, solution]

pdeWithConditions = {D[u[x, t], {t, 2}] - D[u[x, t], {x, 2}] == -9.80665,
    u[x, 0] == 10 x^2 (1 - x)^2,
    Derivative[0, 1][u][x, 0] == 0,
    u[0, t] == 0,
    u[1, t] == 0
}

solution = NDSolve[pdeWithConditions, u[x, t], {x, 0, 1}, {t, 0, 4},
    PrecisionGoal -> 3
]

Plot3D[u[x, t] /. solution, {x, 0, 1}, {t, 0, 4},
    AxesLabel -> {"x", "t", "u"},
    PlotLabel -> "Solution surface of the given Wave equation",
    ColorFunction -> (Blend[{Blue, Cyan, Magenta}, #3] &),
    ColorFunctionScaling -> True
]
```

Out…

```
{u^(0,2)[x, t] - u^(2,0)[x, t] == -9.80665, u[x, 0] == 10 (1 - x)^2 x^2, u^(0,1)[x, 0] == 0, u[0, t] == 0, u[1, t] == 0}
```

```
{{u[x, t] → InterpolatingFunction[ Domain: {{0., 1.}, {0., 4.}} Output: scalar ][x, t]}}
```



Result: The solution of the given Wave equation is obtained and visualized as a surface.

Problem-3: To obtain the solution of the wave equation

$$u_{tt} - u_{xx} = 0, \quad 0 < x < 2, \quad t > 0$$

for the initial conditions

$$u(x,0) = \sin\left(\frac{\pi x}{2}\right), \quad u_t(x,0) = 0, \quad u(0,t) = u(2,t) = 0, \quad 0 \le x \le 2, \quad t \ge 0$$

Solution:

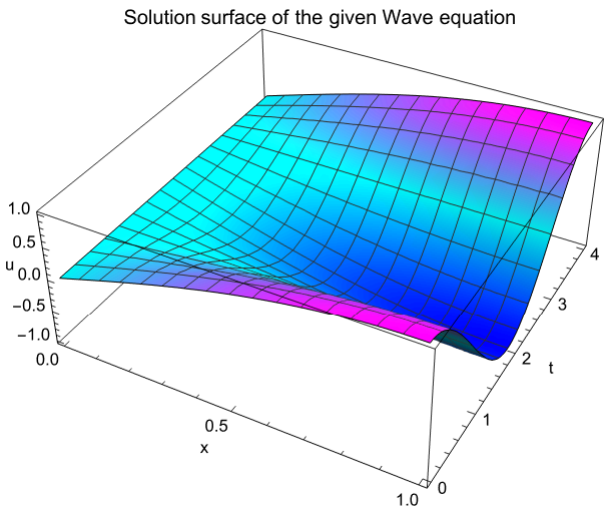
In …

```
Clear[x, t, u, pdeWithConditions, solution]

pdeWithConditions = {D[u[x, t], {t, 2}] - D[u[x, t], {x, 2}] == 0,
    u[x, 0] == Sin[(π x)/2],
    Derivative[0, 1][u][x, 0] == 0,
    u[0, t] == 0,
    u[2, t] == 0
}

solution = DSolve[pdeWithConditions, u[x, t], {x, t}
]

Plot3D[u[x, t] /. solution, {x, 0, 1}, {t, 0, 4},
    AxesLabel -> {"x", "t", "u"},
    PlotLabel -> "Solution surface of the given Wave equation",
    ColorFunction -> (Blend[{Blue, Cyan, Magenta}, #3] &),
    ColorFunctionScaling -> True
]
```

Out…

$$\left\{u^{(0,2)}[x,t]-u^{(2,0)}[x,t]==0,\ u[x,0]==\text{Sin}\left[\frac{\pi x}{2}\right],\ u^{(0,1)}[x,0]==0,\ u[0,t]==0,\ u[2,t]==0\right\}$$

$$\left\{\left\{u[x,t]\to\text{Cos}\left[\frac{\pi t}{2}\right]\text{Sin}\left[\frac{\pi x}{2}\right]\right\}\right\}$$



Result: The solution of the given Wave equation is obtained and visualized as a surface.

Problem-4: To obtain the solution of the wave equation

$$u_{tt} - u_{xx} = 0, \quad 0 < x < 1, \quad t > 0$$

for the initial conditions

$$u(x,0) = \log(1+x^2), \quad u_t(x,0) = 2, \quad 0 \le x \le 1, \quad t \ge 0$$

Solution:

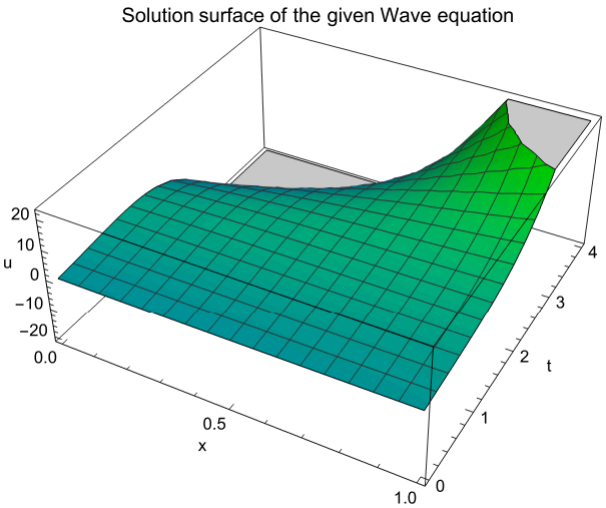
In …

```
Clear[x, t, u, pdeWithConditions, solution]

pdeWithConditions = {D[u[x, t], {t, 2}] - D[u[x, t], {x, 2}] == 0,
    u[x, 0] == Log[1 + x^2],
    Derivative[0, 1][u][x, 0] == 2
}

solution = NDSolve[pdeWithConditions, u[x, t], {x, 0, 1}, {t, 0, 4},
    PrecisionGoal -> 3
]

Plot3D[u[x, t] /. solution, {x, 0, 1}, {t, 0, 4},
    AxesLabel -> {"x", "t", "u"},
    PlotLabel -> "Solution surface of the given Wave equation",
    ColorFunction -> (Blend[{Blue, Green}, #3] &),
    ColorFunctionScaling -> True
]
```

```
NDSolve::bcart: Warning: an insufficient number of boundary conditions have been specified for t
he direction of independent variable x. Artificial boundary effects may be present in the soluti
on.
```

Out…

```
{u^(0,2)[x, t] - u^(2,0)[x, t] == 0, u[x, 0] == Log[1 + x^2], u^(0,1)[x, 0] == 2}
{{u[x, t] → InterpolatingFunction[ Domain: {{0., 1.}, {0., 4.}} Output: scalar ][x, t]}}
```



Result: The solution of the given Wave equation is obtained and visualized as a surface.

Practical-5

Solving systems of Ordinary Differential Equations (ODE)

Command: `DSolve[eqn, y[x], x]`
solves a differential equation for `y[x]`
Command: `DSolve[{eqn1, eqn2, eqn3, ...}, {y1[x], y2[x], y3[x], ...}, x]`
solves a system of differential equation for `yi[x]` where $i = 1, 2, 3, \ldots$

Linear Systems:
The general linear system of two first-order differential equations in two unknown functions x and y is of the form

$$a_1(t)\frac{dx}{dt} + a_2(t)\frac{dy}{dt} + a_3(t)\,x + a_4(t)\,y = F_1(t),$$
$$b_1(t)\frac{dx}{dt} + b_2(t)\frac{dy}{dt} + b_3(t)\,x + b_4(t)\,y = F_2(t) \tag{1}$$

we shall say that a solution of system (1) is an ordered pair of real functions (f, g) such that $x = f(t),\ y = g(t)$ simultaneously satisfy both equations of system (1) on some real interval $a \le t \le b$.

Problem-1: To solve the following system of ordinary differential equations

$$\frac{dx}{dt} + \frac{dy}{dt} - x = -2t,$$
$$\frac{dx}{dt} + \frac{dy}{dt} - 3x = t^2$$

Solution:

In …

```
Clear[x, y, t,
    systems, solution, paricularSol]
```

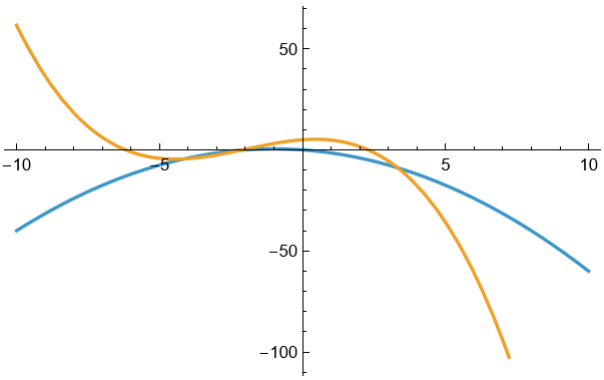
```
systems = {
    x'[t] + y'[t] - x[t] == -2 t,
    x'[t] + y'[t] - 3 x[t] == t^2
}

solution = DSolve[systems, {x, y}, t]

particularSol = {x[t], y[t]} /. solution /. {C[1] -> 5}
Plot[Evaluate[particularSol], {t, -10, 10}]
```

Out…

$$\{-x[t] + x'[t] + y'[t] == -2\,t,\ -3\,x[t] + x'[t] + y'[t] == t^2\}$$

$$\left\{\left\{x \to \text{Function}\left[\{t\}, \frac{1}{2}\left(-2\,t - t^2\right)\right],\ y \to \text{Function}\left[\{t\}, \frac{1}{2}\left(2\,t + t^2\right) + \frac{1}{2}\left(-3\,t^2 - \frac{t^3}{3}\right) + c_1\right]\right\}\right\}$$

$$\left\{\left\{\frac{1}{2}\left(-2\,t - t^2\right),\ 5 + \frac{1}{2}\left(2\,t + t^2\right) + \frac{1}{2}\left(-3\,t^2 - \frac{t^3}{3}\right)\right\}\right\}$$



Result: The solution of the given system is obtained in the output and visualized as a graph.

Problem-2: To solve the following system of ordinary differential equations

$$\frac{dx}{dt} + \frac{dy}{dt} + 4y = \sin(t),$$

$$\frac{dx}{dt} + \frac{dy}{dt} - x - y = 0$$

Solution:

In …

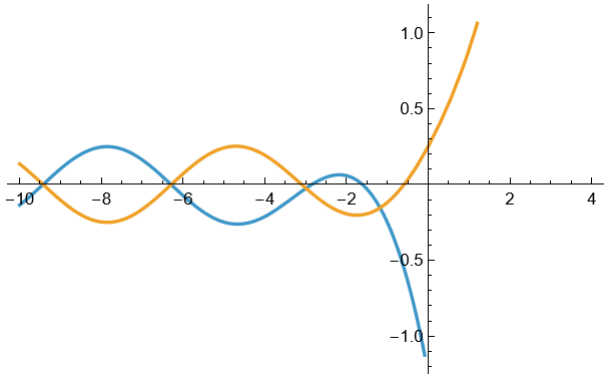
```
Clear[x, y, t,
    systems, solution, paricularSol]

systems = {
    x'[t] + y'[t] + 4 y[t] == Sin[t],
    x'[t] + y'[t] - x[t] - y[t] == 0
}

solution = DSolve[systems, {x, y}, t]

particularSol = {x[t], y[t]} /. solution /. {C[1] -> -1}
Plot[Evaluate[particularSol], {t, -10, 4}]
```

Out…

$$\{4\,y[t] + x'[t] + y'[t] == \mathrm{Sin}[t],\ -x[t] - y[t] + x'[t] + y'[t] == 0\}$$

$$\left\{\left\{x \to \mathrm{Function}\left[\{t\}, \frac{5\,e^{t}\,c_1}{4} - \frac{\mathrm{Sin}[t]}{4}\right],\ y \to \mathrm{Function}\left[\{t\}, -\frac{e^{t}\,c_1}{4} + \frac{\mathrm{Sin}[t]}{4}\right]\right\}\right\}$$

$$\left\{\left\{-\frac{5\,e^{t}}{4} - \frac{\mathrm{Sin}[t]}{4},\ \frac{e^{t}}{4} + \frac{\mathrm{Sin}[t]}{4}\right\}\right\}$$



Result: The solution of the given system is obtained in the output and visualized as a graph.

Problem-3: To solve the following system of ordinary differential equations

$$\frac{dx}{dt} + \frac{dy}{dt} - 2x - 4y = e^{t},$$

$$\frac{dx}{dt} + \frac{dy}{dt} - y = e^{4t}$$

Solution:

In …

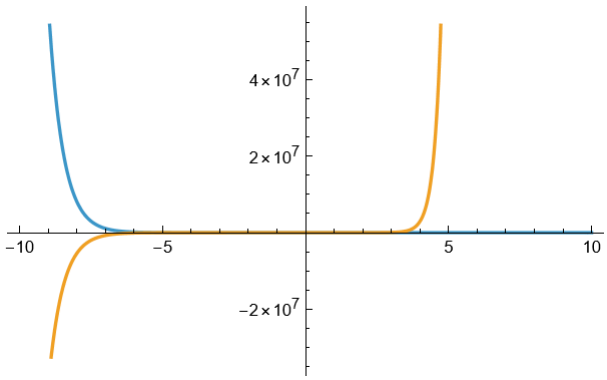
```
Clear[x, y, t,
    systems, solution, paricularSol]

systems = {
    x'[t] + y'[t] - 2 x[t] - 4 y[t] == Exp[t],
    x'[t] + y'[t] - y[t] == Exp[4 t]
}

solution = DSolve[systems, {x, y}, t]

particularSol = {x[t], y[t]} /. solution /. {C[1] -> 2}
Plot[Evaluate[particularSol], {t, -10, 10}]
```

Out…

$$\{-2\,x[t] - 4\,y[t] + x'[t] + y'[t] == e^{t},\ -y[t] + x'[t] + y'[t] == e^{4\,t}\}$$

$$\left\{\left\{x \to \mathrm{Function}\left[\{t\}, -e^{t}\left(-1 + e^{3\,t}\right) + \frac{1}{3}\left(\frac{3}{4}\,e^{-2\,t}\left(-1 + 2\,e^{3\,t}\right)^{2} + e^{-2\,t}\,c_1\right)\right],\ y \to \mathrm{Function}\left[\{t\}, e^{t}\left(-1 + e^{3\,t}\right) - \frac{2}{9}\left(\frac{3}{4}\,e^{-2\,t}\left(-1 + 2\,e^{3\,t}\right)^{2} + e^{-2\,t}\,c_1\right)\right]\right\}\right\}$$

$$\left\{\left\{-e^{t}\left(-1 + e^{3\,t}\right) + \frac{1}{3}\left(2\,e^{-2\,t} + \frac{3}{4}\,e^{-2\,t}\left(-1 + 2\,e^{3\,t}\right)^{2}\right),\ e^{t}\left(-1 + e^{3\,t}\right) - \frac{2}{9}\left(2\,e^{-2\,t} + \frac{3}{4}\,e^{-2\,t}\left(-1 + 2\,e^{3\,t}\right)^{2}\right)\right\}\right\}$$



Result: The solution of the given system is obtained in the output and visualized as a graph.

Practical-6

Solution of One-Dimensional Heat equation $u_t = k\, u_{xx}$ for a homogeneous rod of length l

i.e. To solve the IBVP (Initial Boundary Value Problem):

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2},\ 0 < x < l,\ t > 0$$

$$u(0,t) = 0,\ u(l,t) = 0,\ t \geq 0$$

$$u(0,t) = f(x),\ 0 \leq x \leq l$$

Problem-1: To obtain the solution of the heat equation

$$u_t - u_{xx} = 0,\ \ 0 < x < 5,\ \ t > 0$$

for the initial conditions

$$u(x,0) = 0,\ \ u(0,t) = \sin(t),\ \ u(5,t) = 0,\ \ 0 \leq x \leq 5,\ \ t \geq 0$$

Solution:

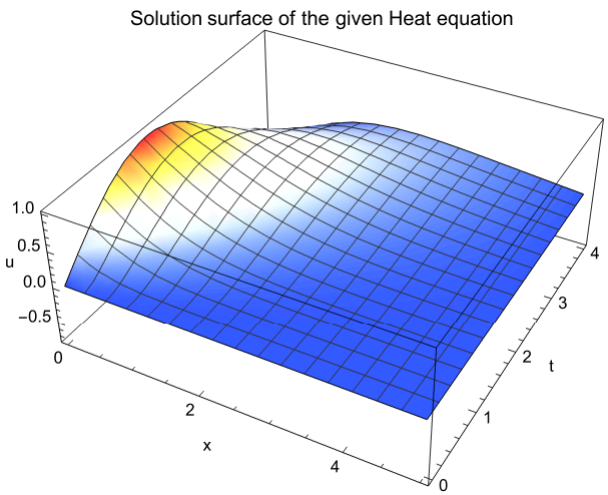
In …

```
Clear[x, t, u, pdeWithConditions, solution]

pdeWithConditions = {D[u[x, t], t] - D[u[x, t], {x, 2}] == 0,
    u[x, 0] == 0,
    u[0, t] == Sin[t],
    u[5, t] == 0
}

solution = NDSolve[pdeWithConditions, u[x, t], {x, 0, 5}, {t, 0, 10},
    PrecisionGoal -> 3
]

Plot3D[u[x, t] /. solution, {x, 0, 5}, {t, 0, 4},
    AxesLabel -> {"x", "t", "u"},
    PlotLabel -> "Solution surface of the given Heat equation",
    ColorFunction -> (ColorData["TemperatureMap"][#3] &),
    ColorFunctionScaling -> False
]
```

Out…

```
{u^(0,1)[x, t] - u^(2,0)[x, t] == 0, u[x, 0] == 0, u[0, t] == Sin[t], u[5, t] == 0}
```

```
{{u[x, t] → InterpolatingFunction[ Domain: {{0., 5.}, {0., 10.}} Output: scalar ][x, t]}}
```



Result: The solution of the given Heat equation is obtained and visualized as a surface.

Problem-2: To obtain the solution of the heat equation

$$u_t - u_{xx} = 0, \quad 0 < x < 10, \quad t > 0$$

for the initial conditions

$$u(x,0) = \tanh(x), \quad u(0,t) = t, \quad u(10,t) = 0, \quad 0 \le x \le 10, \quad t \ge 0$$

Solution:

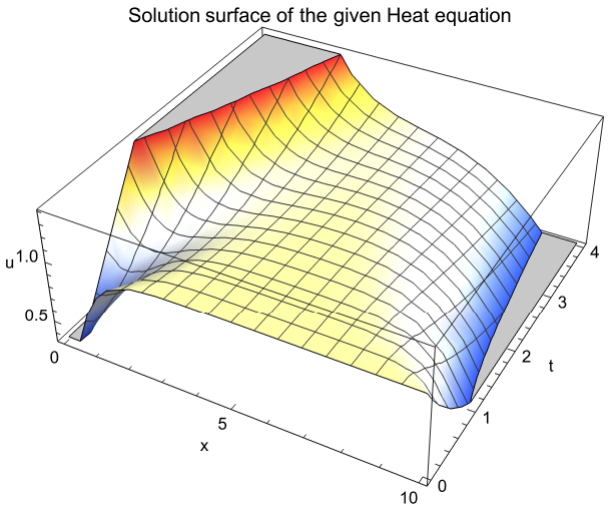
In …

```
Clear[x, t, u, pdeWithConditions, solution]

pdeWithConditions = {D[u[x, t], t] - D[u[x, t], {x, 2}] == 0,
    u[x, 0] == Tanh[x],
    u[0, t] == t,
    u[10, t] == 0
}

solution = NDSolve[pdeWithConditions, u[x, t], {x, 0, 10}, {t, 0, 10},
    PrecisionGoal -> 3
]

Plot3D[u[x, t] /. solution, {x, 0, 10}, {t, 0, 4},
    AxesLabel -> {"x", "t", "u"},
    PlotLabel -> "Solution surface of the given Heat equation",
    ColorFunction -> (ColorData["TemperatureMap"][#3] &),
    ColorFunctionScaling -> True
]
```

```
NDSolve::ibcinc: Warning: boundary and initial conditions are inconsistent.
```

Out…

```
{u^(0,1)[x, t] - u^(2,0)[x, t] == 0, u[x, 0] == Tanh[x], u[0, t] == t, u[10, t] == 0}
{{u[x, t] → InterpolatingFunction[ Domain: {{0., 10.}, {0., 10.}} Output: scalar ][x, t]}}
```



Result: The solution of the given Heat equation is obtained and visualized as a surface.

Problem-3: To obtain the solution of the heat equation

$$u_t - u_{xx} = 0, \quad 0 < x < 5, \quad t > 0$$

for the initial conditions

$$u(x,0) = \sin(x), \quad u(0,t) = 1, \quad u(5,t) = 0, \quad 0 \le x \le 5, \quad t \ge 0$$

Solution:

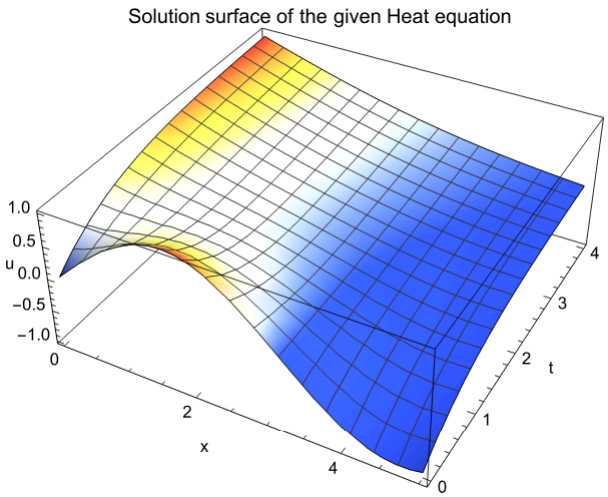
In …

```
Clear[x, t, u, pdeWithConditions, solution]

pdeWithConditions = {D[u[x, t], t] - D[u[x, t], {x, 2}] == 0,
    u[x, 0] == Sin[x],
    u[0, t] == 1,
    u[5, t] == 0
}

solution = NDSolve[pdeWithConditions, u[x, t], {x, 0, 5}, {t, 0, 4},
    PrecisionGoal -> 3
]

Plot3D[u[x, t] /. solution, {x, 0, 5}, {t, 0, 4},
    AxesLabel -> {"x", "t", "u"},
    PlotLabel -> "Solution surface of the given Heat equation",
    ColorFunction -> (ColorData["TemperatureMap"][#3] &),
    ColorFunctionScaling -> False
]
```

```
NDSolve::ibcinc: Warning: boundary and initial conditions are inconsistent.
```

Out…

```
{u^(0,1)[x, t] - u^(2,0)[x, t] == 0, u[x, 0] == Sin[x], u[0, t] == 1, u[5, t] == 0}

{{u[x, t] → InterpolatingFunction[ Domain: {{0., 5.}, {0., 4.}} Output: scalar ][x, t]}}
```



Result: The solution of the given Heat equation is obtained and visualized as a surface.